

SEMESTER END EXAMINATIONS - JULY / AUGUST 2025

Program	: B.E. - Information Science and Engineering	Semester	: IV
Course Name	: Advanced Java	Max. Marks	: 100
Course Code	: 23IS45	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- List the properties of static data member and static method. Create a class called "Employee" with the following data members id, name and address. Using a constructor, initialize these data members. The class also has a static data member named "total_Employees" and static method getTotalEmployees() to display the static data member. Use display() method to display the details of Employee. Create an array of Employee objects and invoke getTotalEmployees() to determine the total number of Employees. CO1 (08)
 - Define abstract class. Is it legal to declare a class as abstract and final? Justify. CO1 (05)
 - Create a class named "BankAccount" with data members accID and balance. With a constructor, initialize the data members. Use withdraw(double amount) method to withdraw money from the account. When the amount to be withdrawn exceeds the balance, use try-catch-finally blocks to handle the exceptions. CO1 (07)
- Define constructor. Mention the special features of constructors. Create a class named "Car" with the following data members namely make, model, year and mileage. Use a default constructor and parameterized constructor to initialize the values of data members. Use a display() method to display the data members of the class. CO1 (08)
 - Define inheritance. Create a class called "Shape" with data member "color" and display() method to display the data member. There are two classes "Rectangle" and "Square" which are subclasses of "Shape". While Rectangle class has data members "length" and "width", Square class has "side" as data member. Both these classes override the "display()" of "Shape" class and the display() of the subclasses compute and display the area. CO1 (07)
 - Define interface. Give the syntax of interface. Differentiate between class and interface. CO1 (05)

UNIT - II

- Explain briefly three methods used to extract characters from String object with example. CO2 (06)
 - Create a Book class with bookId, title, and authorName. Use constructors to initialize the data members. Write a main class to create an ArrayList of books. Display all books details. Program should accept an author name as input and display all books by that author. CO2 (08)

- c) Discuss the following with examples: CO2 (06)
 - i. equals() and equalsIgnoreCase().
 - ii. startsWith() and endsWith().
- 4. a) Differentiate Iterator and ListIterator with suitable examples. CO2 (06)
- b) Create a Student class with rollNo, name, and marks. Use constructors to initialize the data members. Add multiple students to a LinkedList and remove students who scored less than 35. Additionally, include a method to calculate the average marks of all students and display students with marks greater than the average. CO2 (08)
- c) Discuss the benefits of using StringBuffer over String class. Explain getChars(), insert() methods with examples. CO2 (06)

UNIT - III

- 5. a) Elaborate on the two ways of creating a thread with suitable Java code snippets. CO3 (08)
- b) Create a class called Student with data members name, age and marks. Implement a Comparator interface that can sort by marks in descending order. Create an ArrayList of Student objects and use the Collections.sort() method along with the corresponding Comparator to sort the ArrayList based on the criteria. After sorting, print the sorted list of students. CO3 (07)
- c) With suitable Java code snippets, explain Lambda expression and Functional Interfaces. CO3 (05)
- 6. a) What is a HashMap? Give any two constructors of HashMap with syntax. Write a Java program to create a HashMap with USN of students as key and CGPA as value. Populate the HashMap with details of five students and display the contents of HashMap. Are the contents displayed in same order of insertion? CO3 (08)
- b) Create a generic class Complex<T> to represent a complex number with real and imaginary parts. Both the real and imaginary parts can be Integer, Double, or any other numeric type. Use a constructor to initialize the complex number with the real and imaginary parts. Use a method display() that prints the complex number. Create two instances of the Complex class with different numeric types and display the values using the display() method. CO3 (07)
- c) Bring out the purpose of isAlive() and join() methods in Multi-Threading with simple Java code snippets. CO3 (05)

UNIT- IV

- 7. a) Discuss the following swing components with example: CO4 (06)
 - i) JLabel.
 - ii) JTextField.
- b) Explain the architecture of JDBC with key components. CO4 (06)
- c) Write a desktop application where the user can enter employee details such as ID, name, age, department, designation, and salary. Perform validations on all the fields. Display appropriate messages in pop up boxes to indicate wrong entries. On "Add" click, validate and store in a collection. On "Show All" click, display all records in a text area. CO4 (08)

8. a) Explain the usage of radio buttons, check boxes in java swings with suitable example. Write code to indicate an action when a checkbox is selected. CO4 (10)
- b) What is JDBC? Explain the different JDBC driver types. CO4 (10)

UNIT - V

9. a) What is a Java Servlet? Describe the life cycle of a Servlet. CO5 (06)
- b) Write a JSP page that calculates and displays the factorial of a number entered by the user. CO5 (06)
- c) Explain the concept of form data handling in a servlet. How do you set and retrieve session attributes in a servlet? CO5 (08)
10. a) Illustrate the life cycle of a Java Server Pages(JSP) and give example code to demonstrate each component. CO5 (06)
- b) Explain different JSP tags with a program to demonstrate all tags. CO5 (06)
- c) Create a JSP based mini servlet web application that includes a HTML form to collect user details (name, email, phone, gender, address) and a Servlet that processes the form input and displays it on a response page using HTML. CO5 (08)
